# Post-work questionnaire

Laboratory assignment “Battery lifetime prediction”, Embedded Systems (T170B417), Kaunas University of Technology

*The questionnaire is anonymous. Mark one answer for each question unless stated otherwise; in the rating tables, mark one score in each row.*

**Q1. Was the work objective clear from the beginning?**

- ☐ Clear from the beginning
- ☐ Understood only during the work
- ☐ Did not fully understand

**Q2. Which subjects’ skills were mandatory to complete the task?** *(several answers are allowed)*

- ☐ Microcontroller basics (STM32, peripherals, HAL)
- ☐ C programming language
- ☐ Measurement engineering
- ☐ Signal theory
- ☐ Electronic-circuit theory
- ☐ Other: _____

**Q3. Assess the extent to which this work helped develop each of the skills listed below** *(1 = none, 5 = high)*

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Q3.1 Use of Finite State Machines for Embedded System Software Design | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q3.2 Utilization of microcontroller power-saving modes | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q3.3 Estimation of microcontroller current consumption using simulator (CubeMX PCC) | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q3.4 Experimental measurement of the current consumed (X-NUCLEO-LPM01A) | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q3.5 Prediction of battery discharge time | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q3.6 Measurement of program timing (using Keil Event Recorder) | ☐ | ☐ | ☐ | ☐ | ☐ |

**Q4. Assess the complexity of each part of the work** *(1 = easy, 5 = challenging)*

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Q4.1 Design and implementation of a finite state machine (FSM) for implementation of a program algorithm and a specific task case. | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q4.2 Reading digital and analog sensor values | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q4.3 Measurement of program timing (using Keil Event Recorder) | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q4.4 Estimation of microcontroller current consumption using simulator (CubeMX PCC) | ☐ | ☐ | ☐ | ☐ | ☐ |

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Q4.5 Experimental measurement of the current consumed (X-NUCLEO-LPM01A) | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q4.6 Prediction of battery discharge time | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q4.7 Implementation of additional tasks | ☐ | ☐ | ☐ | ☐ | ☐ |

**Q5. To get started, you were provided with a relatively substantial project template (example), the functionality of which you extended according to the task requirements. Do you think you would have developed the designing skills better if you had been required to create it from scratch?**

- ☐ Yes
- ☐ No

**Q6. Did you complete additional tasks to achieve a higher grade?**

- ☐ Yes
- ☐ Time was insufficient
- ☐ Was not interested

**Q7. Effort is proportional to knowledge gained and received grade**

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Your rating | ☐ | ☐ | ☐ | ☐ | ☐ |

**Q8. How many laboratory sessions would be sufficient to complete the work to a high standard (one session corresponds to 2 academic hours)?**

- ☐ 4
- ☐ 6
- ☐ 8
- ☐ More than 8

**Q9. Do you agree with the following statements** *(1 = disagree, 5 = agree)*

| | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Q9.1 There is a lack of knowledge and skills from related adjacent disciplines | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q9.2 A significant disparity in competence and motivation among students, making it difficult for the lecturer to select an appropriate initial teaching level during the course of instruction. | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q9.3 There are more general learning problems such as a lack of planning skills—leaving everything to the last minute (procrastination) | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q9.4 The task description is detailed and clear. | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q9.5 Did this work broaden your understanding of the subject? | ☐ | ☐ | ☐ | ☐ | ☐ |
| Q9.6 Programming is the biggest barrier for me when learning embedded systems. | ☐ | ☐ | ☐ | ☐ | ☐ |

**Q10. Did you use AI tools for some tasks in the work?**

- ☐ Yes
- ☐ No

**Q11. For which tasks did you seek AI assistance? Multiple answers are allowed. Skip if you chose 'No' in Q10**

- ☐ Code debugging / error fixing
- ☐ Clarifying theoretical concepts (measurement method)
- ☐ Verification and elaboration of calculations (battery discharge time)
- ☐ Code generation
- ☐ Interpreting technical documentation
- ☐ Other: _____

**Q12. How has the use of AI tools affected your learning?**

- ☐ Helped understand the subject more deeply
- ☐ Accelerated, but did not understand some parts
- ☐ No effect on learning quality
- ☐ Did not use AI

**Q13. Does this assignment was exceptional compared to the other laboratory assignments in the course? How?**

- ☐ More challenging
- ☐ More practical / applied
- ☐ Similar to others
- ☐ Less interesting

**Q14. How likely would you recommend this assignment to a friend?** *(1 = absolutely not recommend, 5 = highly recommend)*

| | **1** | **2** | **3** | **4** | **5** |
|---|---|---|---|---|---|
| Your rating | ☐ | ☐ | ☐ | ☐ | ☐ |